

torch.max#

<https://pytorch.org/docs/stable/generated/torch.max.html>

Description:

Returns the maximum value of all elements in the input tensor. `amax/amin` supports reducing on multiple dimensions, `amax/amin` does not return indices. Both `amax/amin` evenly distribute gradients between equal values when there are multiple input elements with the same minimum or maximum value.

Function Signature

```
torch.max(input, *, out=None) → Tensor#
```

The difference between `max/min` and `amax/amin` is:

For `max/min`:

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Keyword

out (tuple, optional) – the result tuple of two output tensors (max, max_indices)

Code Examples

```
>>> a = torch.randn(1, 3)
>>> a
tensor([[ 0.6763,  0.7445, -2.2369]])
>>> torch.max(a)
tensor(0.7445)
```

```
>>> a = torch.randn(4, 4)
>>> a
tensor([[ -1.2360, -0.2942, -0.1222,  0.8475],
        [ 1.1949, -1.1127, -2.2379, -0.6702],
        [ 1.5717, -0.9207,  0.1297, -1.8768],
        [-0.6172,  1.0036, -0.6060, -0.2432]])
>>> torch.max(a, 1)
torch.return_types.max(values=tensor([0.8475, 1.1949, 1.5717,
1.0036]), indices=tensor([3, 0, 0, 1]))
>>> a = torch.tensor([[1.0, 2.0], [3.0, 4.0]])
>>> a.max(dim=1, keepdim=True)
torch.return_types.max(
values=tensor([[2.], [4.]]),
indices=tensor([[1], [1]]))
>>> a.max(dim=1, keepdim=False)
torch.return_types.max(
values=tensor([2., 4.]),
indices=tensor([1, 1]))
```

Notes & Warnings

NOTE

Note The difference between max/min and amax/amin is: amax/amin supports reducing on multiple dimensions, amax/amin does not return indices. Both amax/amin evenly distribute gradients between equal values when there are multiple input elements with the same minimum or maximum value. For max/min: If reduce over all dimensions(no dim specified), gradients evenly distribute between equally max/min values. If reduce over one specified axis, only propagate to the indexed element.

NOTE

Note If there are multiple maximal values in a reduced row then the indices of the first maximal value are returned.

Detailed Documentation

Documentation

Returns the maximum value of all elements in the input tensor.

amax/amin supports reducing on multiple dimensions,

amax/amin does not return indices.

Both amax/amin evenly distribute gradients between equal values when there are multiple input elements with the same minimum or maximum value.

If reduce over all dimensions(no dim specified), gradients evenly distribute between equally max/min values.

If reduce over one specified axis, only propagate to the indexed element.

input (Tensor) – the input tensor.

out (Tensor, optional) – the output tensor.

Returns a namedtuple (values, indices) where values is the maximum value of each row of the input tensor in the given dimension dim. And indices is the index location of each maximum value found (argmax).

If keepdim is True, the output tensors are of the same size as input except in the dimension dim where they are of size 1. Otherwise, dim is squeezed (see `torch.squeeze()`), resulting in the output tensors having 1 fewer dimension than input.

If there are multiple maximal values in a reduced row then the indices of the first maximal value are returned.

input (Tensor) – the input tensor.

dim (int, optional) – the dimension to reduce. If omitted, all dimensions are reduced. Explicit None is not supported.

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

out (tuple, optional) – the result tuple of two output tensors (max, max_indices)